

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background.

A COMPREHENSIVE COVERAGE

MODULE - 5

BASIC CONCEPTS - 5

ECOLOGICAL NICHE

- Niche refers to the unique functional role and position of a species in its habitat or ecosystem.
- In nature, many species occupy the same habitat but they perform different functions.
- Habitat of a species is like its 'address' (i.e. where it lives) whereas niche can be thought of as its "profession" (i.e. activities and responses specific to the species).
- A niche is unique for a species while many species share the habitat. No two species in a habitat can have the same niche. This is because of the competition with one another.
- A species' niche includes all of its interactions with the biotic and abiotic factors of its environment.



Significance of Ecological Niche:

Niche plays an important role in conservation of organisms. If we have to conserve species in its native habitat, we should have knowledge about the niche of the species and should ensure that all requirements of its niche are fulfilled.

MECHANISMS OF ADAPTATION

How do organisms respond to stressful conditions in their habitats?

1 REGULATE

- It means maintaining Homeostasis.
- Some organisms are able to maintain Homeostasis, which ensures constant body temperature or constant osmotic concentration.
- All birds and mammals and few other organisms are capable of such regulation. Plants do not have the mechanism to maintain constant internal temperature.

Homeostasis

- *Few organisms try to maintain constant internal condition despite varying external environmental conditions. This process is called Homeostasis.*
- *This constancy could be in terms of temperature and osmotic concentration.*

MECHANISMS OF ADAPTATION

How do organisms respond to stressful conditions in their habitats?

2 CONFORM

- All the plants and many animals cannot maintain a constant internal condition. Their body temperature changes with the ambient temperature.
- In aquatic organisms, the osmotic concentration of their body fluids changes with that of the ambient water osmotic concentration.
- Some species regulate but over a limited range of environmental conditions, beyond which they simply conform.

Example: Deciduous trees shed their leaves during the dry season.



MECHANISMS OF ADAPTATION

How do organisms respond to stressful conditions in their habitats?

3 MIGRATE

- The organism can move away temporarily from the stressful habitat to a more hospitable habitat and return back when the stressful period is over.
- The Sociable Lapwing migrates to Western India from Central Asia during winter.



MECHANISMS OF ADAPTATION

How do organisms respond to stressful conditions in their habitats?

4 SUSPEND

- In some plants, the seeds germinate under favorable environmental conditions. During unfavorable conditions, they go into the state of dormancy.
- Some animals like polar bear go into hibernation (winter sleep). Some snails and fish go into aestivation (summer sleep) to avoid summer related problems like heat.



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